

MINHYUK JANG

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Education

University of Illinois Urbana-Champaign (UIUC)

Aug. 2025 – PRESENT

Ph.D. Student in Mechanical Science and Engineering - (GPA: 4.0/4.0)

- Research Interests: Robotics, Learning-based control, Machine Learning, Safe autonomy
- Advanced Controls Research Laboratory - Advisor: Prof. Naira Hovakimyan

Seoul National University (SNU)

2019 – 2025

B.S. in Artificial Intelligence and Mechanical Engineering (Robotics Track)

- (GPA : 3.99/4.0), Summa Cum Laude
- College of Engineering Outstanding Graduate Award, Outstanding B.S. Thesis Presentation Award
- 18 months of military service included

Publications

- Minhyuk Jang, Jungjin Lee, Astghik Hakobyan, Naira Hovakimyan, and Insoon Yang. *Residual-Aware Distributionally Robust EKF: Absorbing Linearization Mismatch via Wasserstein Ambiguity*, IEEE Control and Decision Conference (CDC), 2026, under review, [\[arXiv\]](#)
- Minhyuk Jang, Astghik Hakobyan, and Insoon Yang. *Distributionally Robust Kalman Filter*, IEEE Transactions on Automatic Control (TAC), under review, [\[arXiv\]](#)
- Minhyuk Jang, Astghik Hakobyan, and Insoon Yang. *On the Steady-State Distributionally Robust Kalman Filter*, IEEE Control and Decision Conference (CDC), 2025, [\[arXiv\]](#)
- Minhyuk Jang, Astghik Hakobyan, and Insoon Yang. *Distributionally Robust Control and State Estimation for Linear Stochastic Systems*, [\[arXiv\]](#)
- Minhyuk Jang. *Stability Analysis of Disturbance Observer under Model Uncertainty with Different System Degrees between True and Nominal Systems*, Institute of Control, Robotics and Systems (ICROS), 2024

Experience

Advanced Controls Research Laboratory

Aug. 2025 – Present

Research Assistant - Advisor: Naira Hovakimyan

Illinois, United States

- **Current Project:** Proposed a Distributionally Robust Extended Kalman Filter (DR-EKF) to improve estimation performance under noise-distribution mismatch and to account for linearization errors.
- **Current Project:** Developing an energy-aware aerial physical interaction framework for an omnidirectional drone, using robust flight control via $\mathcal{L}_1$ adaptive control.

SNU Learning and Decision Systems (LDS) Lab

Mar. 2023 – Aug. 2025

Research Intern - Advisor: Prof. Insoon Yang

Seoul, South Korea

- Proposed a steady-state Distributionally Robust Kalman Filter that accounts for distributional uncertainties in disturbances and measurement noise; published as first author at IEEE CDC 2025 with an extended version submitted to IEEE TAC
- Developed a unified framework for Distributionally Robust Control and State Estimation, leveraging game-theoretic methods to address uncertainties in disturbance and noise distributions

NEARTHLAB

Aerospace Engineering Intern, GNC Team

Jan. 2024 – Feb. 2024

Seoul, South Korea

- Implemented DOB and LQR based position/velocity controllers in C++&ROS2 and tested through Gazebo simulation
- Integrated a flight controller with a companion computer for quadrotor offboard control, implemented DOB+LQR and DOB+PID trajectory tracking controllers, and conducted extensive outdoor flight experiments

Teaching Experience & Professional Service

- **Teaching Experience:** Teaching Assistant (TA), UIUC TAM 210/211, Statics (Spring 2026).
- **Professional Service:** Reviewer, *IEEE Transactions on Signal Processing*.

Selected Projects (See more at [HERE](#))

Automated Hyperparameter Tuning Algorithm for MPPI Control

Sep. 2023 – Jun. 2024

Outstanding B.S. Thesis Presentation Award

Seoul, South Korea

- Designed an adaptive algorithm to automatically adjust the hyperparameter (Inverse temperature) for MPPI control
- Implemented the algorithm in the MuJoCo MPC (MJPC) framework, improving control performance, reducing state fluctuations and control costs, and enabling smoother trajectories in quadrotor hovering and racing tasks

VTOL (Vertical Take-off and Landing) Projects

Mar. 2024 – Sep. 2024

Team Leader – Korea Robot Aircraft Competition (Grand Award)

Taeon, South Korea

- Constructed two A-tail Quadplane VTOLs with autonomous flight capabilities, each with wingspans of 1.5m and 2.0m
- Oversaw the full system engineering process, from selecting electronic components (servos, motors, flight controller, batteries, GPS, PDB, RC, etc.) to wiring, sensor calibration, and mission task planning

Leadership / Extracurricular activities

Bulnabi - SNU Drone Club

Sep. 2023 – Feb. 2025

Team Leader

Seoul, South Korea

- Organized and conducted over five Quadrotor Build/Fly seminars, teaching hardware assembly, sensor calibration, Ground Control Station usage, flight experiments, and flight log analysis
- Led a 25-member team for the Korea Robot Aircraft Competition, focusing on VTOL system design and autonomous flight missions; conducted over 30 outdoor fixed-wing missions and successfully stabilized all flight phases.

KATUSA (Korean Augmentation to the United States Army)

Sep. 2021 – Mar. 2023

Squad Leader – Sergeant, 8th Army, Camp Humphreys

Pyeongtaek, South Korea

- Led and managed a 10-soldier squad, ensuring their training, well-being, and mission preparedness
- Provided essential translation and interpretation support during Combined Exercises, leveraging language proficiency
- Engaged in daily collaboration with American colleagues within a U.S. Army office

G.I.V - SNU Volunteering Club

Mar. 2019 – Sep. 2020

Vice President

Seoul, South Korea

- Delivered educational support in math, science, and art to teenagers in rural areas, emphasizing experiential learning
- Organized and engaged in various volunteer activities, including secondhand markets, mural painting, and food drives

Technical Skills

Programming: C/C++, Python, MATLAB (Simulink)

Libraries/Softwares: PX4-Autopilot, VICON, ROS2, Gazebo, Arduino, MuJoCo, PyTorch, Fusion 360, L^AT_EX

Hardware: Robotic system prototyping and testing (aerial vehicles, mobile robots), VTOL development (autonomous flight missions), Quadrotor and hexacopter operation, FPV drone piloting, Crazyflie, electronics integration (wiring, soldering), fabrication (laser cutting, 3D printing)

Languages: Korean (native), English (TOEFL iBT 113/120)

Honors / Awards

College of Engineering Outstanding Graduate Award <i>One of the Top 24 Graduates across the entire College of Engineering</i>	2025 <i>SNU College of Engineering Alumni Association</i>
Grand Award - Korea Robot Aircraft Competition <i>1st Place among 39 University Teams</i>	2024 <i>Minister of Commerce, Industry and Energy</i>
Outstanding B.S. Thesis Presentation Award <i>Department of Mechanical Engineering</i>	2024 <i>Seoul National University</i>
Outstanding Materials and Manufacturing Processes Award <i>Materials and Manufacturing Processes Contest</i>	2024 <i>Seoul National University</i>
Excellence Award - Mechatronics Competition <i>Department of Mechanical Engineering</i>	2023 <i>Seoul National University</i>
ARCOM (Army Commendation Medal) <i>United States Department of the Army</i>	2022 <i>Camp Humphreys</i>
Kwanjeong Scholarship <i>Recipient of a full tuition scholarship along with a stipend for two years</i>	2021 <i>Kwanjeong Educational Foundation</i>
Merit-based Scholarship <i>Department of Mechanical Engineering</i>	Spring 2020, Fall 2020 <i>Seoul National University</i>

Relevant Coursework

Control: Nonlinear System Theory, Robust Adaptive Control, Advanced Control Techniques, Mechanical System Modeling and Control

Machine Learning: MDP and Reinforcement Learning, Basics of Deep Learning, Artificial Intelligence, Machine Learning Fundamentals and Applications in Electrical and Computer Engineering, Machine Listening

Robotics: Autonomous Vehicle Systems Engineering, Introduction to Robotics, Mechanical Product Design, Mechatronics, Materials and Manufacturing Processes

Talks

- **Invited Talks:** *Are you confident your sensor is tuned correctly? – Robust Kalman filtering under noise uncertainty*, Korean-American Robotics Association (KARA), 2026
- **Invited Talks:** *Design and Development of a Lightweight, 3D-Printed VTOL Aircraft with Autonomous Flight Capabilities*, Center for Scientific Innovation and Education (CSIE), Yerevan, Armenia, 2024 (online)
- **Invited Talks:** *Design and Development of a Lightweight, 3D-Printed VTOL Aircraft with Autonomous Flight Capabilities for Advanced Air Mobility (AAM)*, Unmanned Vehicle Systems Symposium, Daejeon, South Korea, 2024